

SYNOPSIS

GRIDSENSE AI – AI-POWERED SMART GRID, ENERGY DEMAND & ANOMALY INTELLIGENCE PLATFORM

Praveen M | Id - AF05305017

ABOUT THE TOPIC (DATASET)

GridSense AI is a Power BI-based smart energy analytics platform designed to explore electricity consumption, solar generation, demand patterns, weather conditions, and anomaly indicators. It organizes smart-grid meter data into interactive report pages to help users review energy usage, identify flagged readings, examine forecast-related values, and explore relationships between environmental conditions and energy demand.

The project uses a cleaned smart-grid dataset containing approximately 20,000 records. Each row represents a meter observation at a recorded timestamp. The dashboard contains four pages: Energy Overview, AI Energy & Anomaly Intelligence, Energy Forecasting & Consumption Insights, and Weather & Environmental Intelligence.

DATA STRUCTURE

The dataset is organized in tabular format. Important columns include:

- **timestamp / date / month / day_name:** Time fields for trends and periodic consumption patterns.
- **meter_id / region / building_type:** Meter, geographic region, and building category.
- **consumption_kwh / solar_generation_kwh:** Energy consumed and solar energy generated.
- **next_hour_consumption_kwh / lag1_kwh / lag24_kwh:** Forecast-related and historical consumption values.
- **temp_c / wind_ms / humidity_pct:** Weather conditions for comparison with energy consumption and solar generation.
- **anomaly_flag / high_usage_flag / sensor_health:** Flagged anomalies, high-usage events, and sensor status.
- **outage_risk_score / risk_category:** Risk-related values and categories for dashboard analysis.
- **peak_period / Temperature Range / Forecast Error:** Derived fields for peak periods, temperature bands, and forecast-related differences.

These fields support interactive reporting. A difference between next-hour predicted consumption and current-hour consumption is treated as a comparison, not a verified forecast error, unless the actual next-hour value is correctly matched.

DATA PREPROCESSING

1. **Data Cleaning:** Check missing values, data types, duplicates, and inconsistent entries.
2. **Data Transformation:** Prepare timestamp-derived fields such as date, month, day name, and hour.
3. **Calculated Columns:** Create derived categories such as peak period, temperature range, and forecast-related bands where applicable.
4. **Data Modeling & DAX:** Create measures for consumption, solar generation, anomaly counts, risk score, and weather KPIs.
5. **Validation:** Review totals, category values, slicer behavior, and visual aggregations.

IMPLEMENTATION STEPS

6. **Data Ingestion:** Import the cleaned smart-grid CSV dataset into Power BI Desktop.
7. **Preprocessing:** Clean and transform data using Power Query and create required derived fields.

8. **Data Modeling:** Configure columns, aggregations, and DAX measures for KPIs.
9. **Dashboard Development:** Build four interactive pages with KPI cards, charts, slicers, and page navigation.
10. **Analysis:** Explore consumption trends, anomaly indicators, forecast-related patterns, and weather relationships.
11. **Validation & Reporting:** Test filters and calculations, refine the dashboard, and document insights.

DASHBOARD PAGES

- **Page 1 – Energy Overview:** Consumption KPIs, trends, region/building comparisons, and solar generation versus grid consumption.
- **Page 2 – AI Energy & Anomaly Intelligence:** Anomaly trend, risk-category distribution, high-usage analysis, sensor health, and outage-risk indicators.
- **Page 3 – Energy Forecasting & Consumption Insights:** Actual versus next-hour predicted consumption, hourly/weekly patterns, peak consumption, and peak/off-peak comparisons.
- **Page 4 – Weather & Environmental Intelligence:** Temperature, wind and humidity KPIs; weather-versus-consumption analysis; and solar generation by temperature range.

TECHNOLOGIES USED

Microsoft Power BI Desktop, Power Query, DAX, CSV dataset, and Microsoft Excel (for data inspection or preparation where required).

SOFTWARE REQUIREMENTS

Operating System: Windows 10/11 or a compatible system. Software: Microsoft Power BI Desktop; Microsoft Excel or a compatible spreadsheet tool (optional).

HARDWARE REQUIREMENTS

Minimum: 4 GB RAM and an Intel Core i3-class processor or equivalent. Recommended: 8 GB RAM or higher for smoother report authoring and data handling.

EXPECTED OUTCOME

The expected outcome is an interactive four-page Power BI dashboard that consolidates smart-grid energy and weather-related data into readable KPIs and visual reports. It supports exploratory analysis and decision-making; it does not independently establish causality or guarantee operational outage prediction.